# A transdiagnostic map places immunometabolic burden as a distinct, durable, prognostic axis of severe mental illness

Andrii Kulakovskyi[1*] Ophélia Godin[1,2] Léah Camarcat[1] Engin Altunlu[1] Marion Leboyer[1,2]

[1]Fondation FondaMental [2]Inserm

July 5, 2026

**Abstract**

Psychiatric diagnoses group biologically heterogeneous patients, and dimensional alternatives have struggled to show that any single biological signal is more than a correlate of overall severity. We built **FACE-ATLAS**, a diagnosis-blind dimensional map of $N = 9{,}013$ patients with bipolar disorder, schizophrenia or major depression (three harmonized FACE cohorts, 21 expert-centre sites), in which routinely-collected biology is placed as a *co-equal latent axis* alongside cognition, sleep and symptom dimensions rather than as a downstream correlate. A single global Bayesian bifactor model—fit to each patient's observed cells only (no imputation), with mixed likelihoods, uncertainty propagated end-to-end, and the map frozen before any validation—yields eight axes: a general functional-burden factor ($G$) and seven specific dimensions (cognition, immunometabolic load, sleep, mania/activation, suicidality, developmental risk and substance use). The patient space is a graded **continuum, not discrete biotypes** (apparent clustering is reproduced by a single-Gaussian null), summarized by a five-archetype simplex that locates a patient's clinical–biological state $\approx$ 9.7$\times$ more tightly than the DSM-5 label. Allowing $G$ to correlate freely with the specific axes, one property recurs across every validation stage: an **immunometabolic axis nearly independent of general burden** (sharing $\approx$ 1% of its variance with $G$ versus 15–18% for cognition and sleep—a correlation of $\approx$ 0.10 against 0.39/0.42; robust to medication, adiposity and site), which is the **most temporally durable** dimension (test–retest intraclass correlation 0.91) and the **worst-prognosis pole** for two-year functioning (functional remission 22%–63% across corners). This separation is *relative* and *measured*, and converges with an independent immunometabolic-depression literature. The map improves prediction of future functioning beyond diagnosis and severity at the group level (held-out predictive-density gain +62.8) but not at the individual level (AUC +0.010), positioning the immunometabolic axis as a **cohort-enrichment and monitoring target rather than an individual treatment rule**. All results are internal-validity findings on observational data; individual-level utility would require external validation and incident-event outcomes. Placing routinely-collected biology *inside* the transdiagnostic factor space—as a co-equal latent axis rather than a correlate of symptom dimensions—is, to our knowledge, the first such map across bipolar disorder, schizophrenia and major depression.

## 1 Introduction

Psychiatric diagnoses group biologically heterogeneous patients under a single label, and two patients meeting criteria for the same disorder may share little in their cognition, biology or course. Dimensional frameworks were introduced to replace these categories with continuous

*Correspondence: andrii.kulakovskyi@fondation-fondamental.org

axes that cut across diagnoses [1–3], and a general factor of psychopathology—the tendency for symptoms to co-occur and severity to load together—has become their organizing backbone [4, 5]. A dimensional alternative, however, is only as good as the *measurement* that produces it: to be credible across cohorts it must be comparable between sites, faithful to the mixed data types of a clinical assessment, honest about the missingness that pervades real registries, and stable enough that an axis defined once keeps its meaning when it is later validated.

Where biology has entered these maps, it has almost always entered as a *correlate*—a biomarker regressed on a symptom dimension, or a stratification computed after the fact—rather than as a coordinate of the space itself [6–8]. This choice has a consequence that is easy to miss. If biology is only ever compared to severity, one cannot ask whether a biological axis is *separable* from severity, because the machinery that would answer the question has been assumed away. The immunometabolic burden of severe mental illness—weight, adiposity, dysglycaemia, inflammation—is a natural test case: it is elevated transdiagnostically, tracks medication and chronicity, and has been argued to form a partly distinct dimension of illness [9–14]. Yet whether that dimension stands *apart from* the general severity axis, rather than riding along with it, has been described qualitatively but never measured inside a single transdiagnostic map.

Here we build such a map. We fit one global, missingness-aware Bayesian bifactor model to the harmonized baseline assessment of $N = 9{,}013$ patients with bipolar disorder, schizophrenia or major depression from three national cohorts, with diagnosis withheld from model fitting throughout, and place routinely-collected biology *inside* the factor space as a co-equal latent axis alongside cognition, sleep and the symptom dimensions. We then interrogate the fixed map with a single question asked of every validation stage: *what stands apart, what stays, and what predicts?* The answer converges on one axis. Immunometabolic load is nearly independent of general burden when the model is free to entangle them, is the most temporally durable dimension of the eight, and marks the worst-prognosis pole of the patient space—an "immunometabolic island" that a diagnosis or a severity score does not resolve.

We are explicit about what is and is not new. That immunometabolic biology matters in severe mental illness, and forms a partly separable dimension, is established; our finding *converges* with that independent literature, which is why we report it as relative, measured independence rather than a new biological claim. What is new is the map: to our knowledge the first diagnosis-blind dimensional atlas spanning these three disorders that carries biology as a co-equal axis, and the measurement discipline—no imputation, mixed likelihoods, uncertainty propagated end-to-end, the map frozen before validation—that turns "biology is separable from severity" from a modelling assumption into an estimated quantity.

# 2 Results

## 2.1 A diagnosis-blind map of severe mental illness

We assembled the harmonized baseline assessment of $N = 9{,}013$ patients (bipolar disorder 6,252; schizophrenia 2,209; major depression 552) from three national FACE cohorts across 21 expert centres, spanning a 143-variable pool of cognitive tests, symptom scales, functioning measures, sleep and chronotype instruments, developmental and substance-use history, and a routine biology panel (body composition, lipids, glycaemia, inflammation; cohort assembly and follow-up attrition in Extended Data Fig. E1). A single global Bayesian sparse bifactor / exploratory structural-equation model was fit to each patient's observed cells only—no imputation, mixed likelihoods for continuous, ordinal and count indicators, and a Gaussian-copula block for the correlated biology panel—estimating an eight-dimensional map: a general functional-burden factor ($G$) and seven specific dimensions (cognition, immunometabolic load, sleep, mania/activation, suicidality, developmental risk and substance use). Diagnosis was withheld from fitting throughout; the map is defined by the data, not the label.

The bifactor identification fixes $G$ orthogonal to the specific axes—this orthogonality is an identifying *coordinate choice*, not an empirical claim, and the resulting map (the *primary map*) is the coordinate system on which every downstream analysis is scored. Each axis is anchored by its expected indicators (Figure 1): general burden by global functioning and severity scales, cognition by memory and processing-speed tests, immunometabolic load by body-mass, waist circumference, triglycerides, urate and C-reactive protein, and so on through the symptom and history dimensions. The full 143-indicator loading atlas is given in Extended Data Fig. E2, and measurement invariance across the three cohorts in Extended Data Fig. E3. With the map fixed, we then ask three questions of it in turn—whether it resolves discrete biotypes or a continuum, whether any axis stands apart from general severity, and whether that axis persists and predicts.

## 2.2 The patient space is a continuum, summarized by five archetypes

On the fixed map we asked whether patients fall into discrete biotypes or occupy a graded continuum. An uncertainty-aware structure test found no clustering beyond that expected of a single multivariate-Gaussian blob: the best silhouette (a 0–1 cluster-separation score) over $K = 2$–$12$ clusters (0.140) is within noise of the Gaussian null (0.137, $z = 1.13$), and a Gaussian mixture selects $K = 1$ in 20/20 posterior draws (Figure 2). Projected to two dimensions, the cohort is a single cloud in which the three diagnoses are fully intermixed (adjusted Rand index 0.021, where 0 is chance agreement and 1 a perfect match to the DSM-5 label), general burden varies as a smooth gradient, and—critically—immunometabolic load varies along a *different* direction than general burden, visible as a distinct axis of the same cloud.

Because a continuum still needs a vocabulary, we summarized it with a stable five-archetype simplex: five extreme phenotypes whose convex combinations locate every patient as a blend of weights summing to one. The archetype encoding describes a patient's clinical–biological state $\approx 9.7\times$ more tightly than the DSM-5 diagnosis—it accounts for a mean $\eta^2 = 0.256$ of the between-patient variance in factor scores (where $\eta^2$ is the 0–1 fraction of variance an encoding explains), against 0.026 for the diagnosis and 0.018 for the cohort of origin. The five corners are interpretable—an activation/sleep pole, a severe "clean-biology" pole, an **immunometabolic pole**, a trauma/suicidality pole and a low-burden "well" pole—and two of them, the immunometabolic and the severe clean-biology corners, share high general burden but sit at opposite ends of the biology axis. Whether that contrast reflects a genuine separation of biology from severity, or merely the map's built-in coordinates, is what we test next.

## 2.3 Immunometabolic load separates from general severity (the hinge)

The map's defining property is read here. Because the bifactor fit fixes $G$ orthogonal to the specifics by construction, a domain's entanglement with general burden cannot be read from that model—it has been set to zero by the coordinate choice. To convert "biology is separable from severity" from a modelling constraint into a measured quantity, we re-estimated the model allowing $G$ to correlate freely with the specific axes. The resulting vector of general–specific correlations, $\phi_{Gd} = \mathrm{Corr}(G, D)$, is the *primary estimand* of FACE-ATLAS for the biology–severity question; the bifactor map remains the coordinate system every other analysis is scored on, but the separation of biology from severity is read from $\phi_{Gd}$, not from the map's built-in orthogonality.

Under this correlated-$G$ model, immunometabolic load was the *least* entangled with functional burden of any domain: a correlation with $G$ of $\approx 0.10$, against 0.39 for cognition and 0.42 for sleep (Figure 3). In shared-variance terms the gap is stark—immunometabolic load shares only $\approx 1\%$ of its variance with general burden, against $\approx 15\%$ (cognition) and $\approx 18\%$ (sleep). The separation was robust to the obvious confounders: adjusting each indicator for age, sex, education and recruitment site, and then additionally for antipsychotic exposure, did not raise the immunometabolic correlation—if anything it fell, to $\approx 0.07$—and immunometabolic load

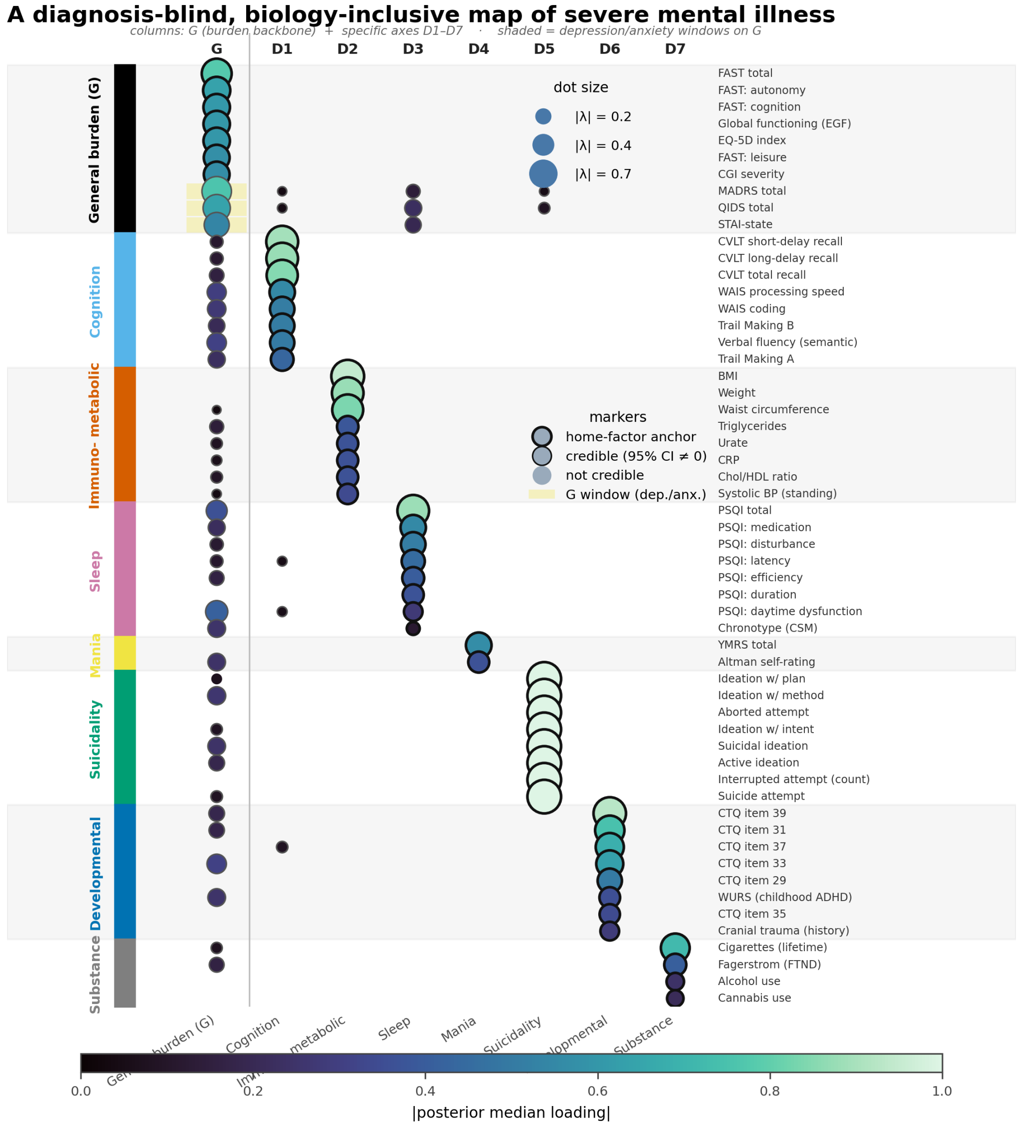


Figure 1: **A diagnosis-blind, biology-inclusive map of severe mental illness.** Posterior indicator×factor loadings for the eight-axis model fit to $N = 9{,}013$ patients' observed cells (no imputation), with the map *frozen before any validation stage*—a pre-specification that guards the downstream claims against overfitting. Routinely-collected biology (immunometabolic panel) enters as a co-equal latent axis alongside cognition, sleep and the symptom dimensions rather than as a downstream correlate. Marker size is |loading|; the general factor $G$ is orthogonal to the specifics by construction (the primary-map coordinate choice). This map is the coordinate system; the degree to which each axis is entangled with general burden is not read from it (that is set to zero here by construction) but estimated from a free-correlation re-fit, where the immunometabolic axis proves the least entangled of all (Fig. 3). Unlike prior single-diagnosis immunometabolic-depression subtyping, biology here is a co-equal axis (not a correlate), measured without imputation, across bipolar disorder, schizophrenia and major depression.

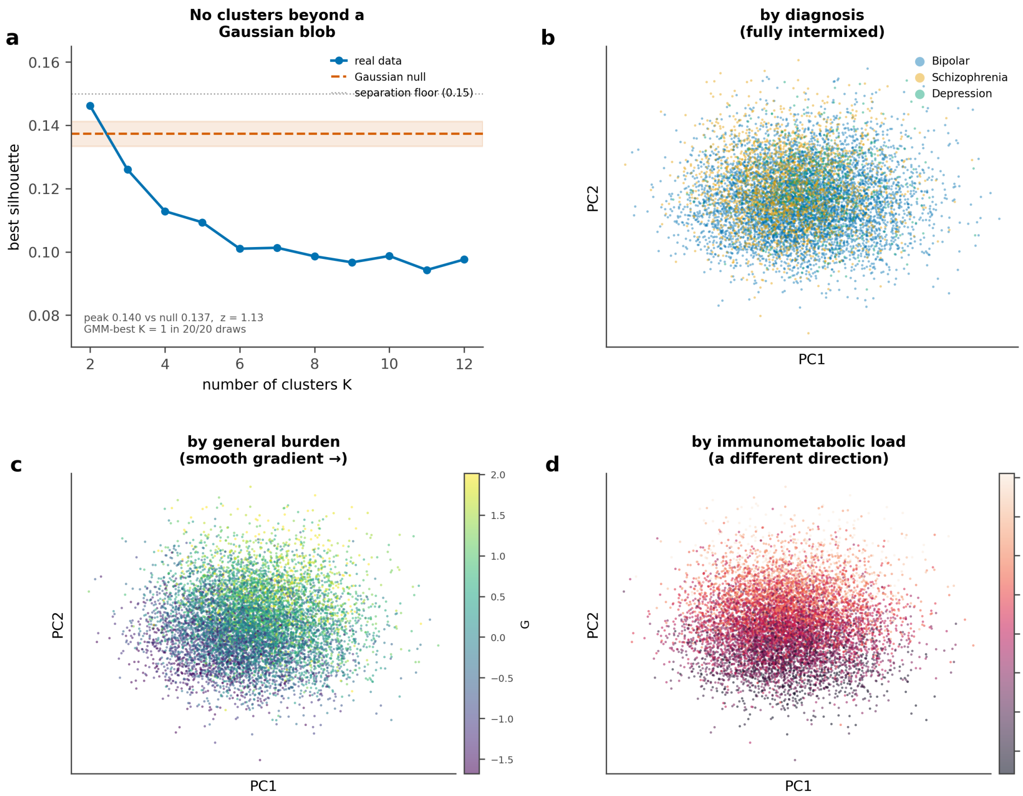


Figure 2: **A graded continuum, not discrete biotypes.** (a) Best silhouette versus a single-Gaussian null across $K$; the real data never clear the separation floor and a mixture selects $K = 1$ in every draw. (b) The two-dimensional cloud with diagnoses fully intermixed. (c) General burden as a smooth gradient. (d) Immunometabolic load varies along a different direction—a separate axis of the same continuum, not a pole of the severity gradient.

remained the least entangled axis at every step (Extended Data Fig. E6). This is *relative*, measured independence, quantifying at the low end of what an independent immunometabolic-depression literature would predict [12–15], now quantified inside one map. Clinically, it means two patients with the same overall impairment can carry very different immunometabolic loads—a difference the global severity picture does not capture.

## 2.4 The separable axis is the durable one

If immunometabolic load is a distinct axis, is it a stable trait or a passing state? Scoring two yearly follow-ups under the frozen map, a between/within/measurement variance decomposition placed immunometabolic load as the most temporally durable of the eight axes: a test–retest intraclass correlation (ICC, the 0–1 fraction of an axis's variance that is stable trait rather than transient state) of 0.91, well above cognition (0.70), general burden (0.62) and every symptom axis (sleep 0.47, suicidality 0.43, developmental risk 0.39; Figure 4). The symptom axes move while the biology holds: over two years the cohort improved on symptoms and severity, yet patients kept their rank on the immunometabolic axis almost exactly. An axis this durable is one a cohort could in principle be enriched or stratified on—it is still there at follow-up—whereas the fast-moving symptom axes suit moment-to-moment monitoring instead.

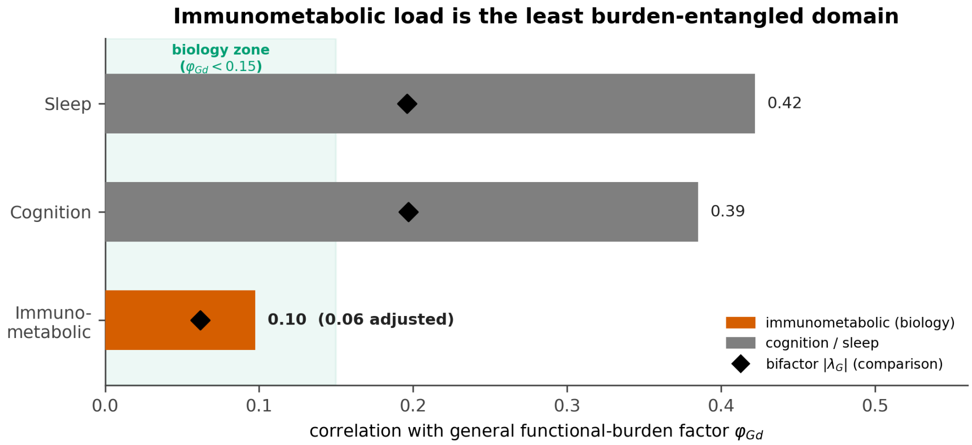


Figure 3: **Immunometabolic load is the least burden-entangled domain.** Free (correlated-$G$) correlation of each specific axis with general burden $\phi_{Gd}$ (bars), with the bifactor direct loadings $|\lambda_G|$ shown for comparison (diamonds). Immunometabolic load sits in the "biology zone" at $\approx 0.10$ (0.06 under confound adjustment), against 0.39/0.42 for cognition/sleep. This is the primary estimand for the biology–severity question.

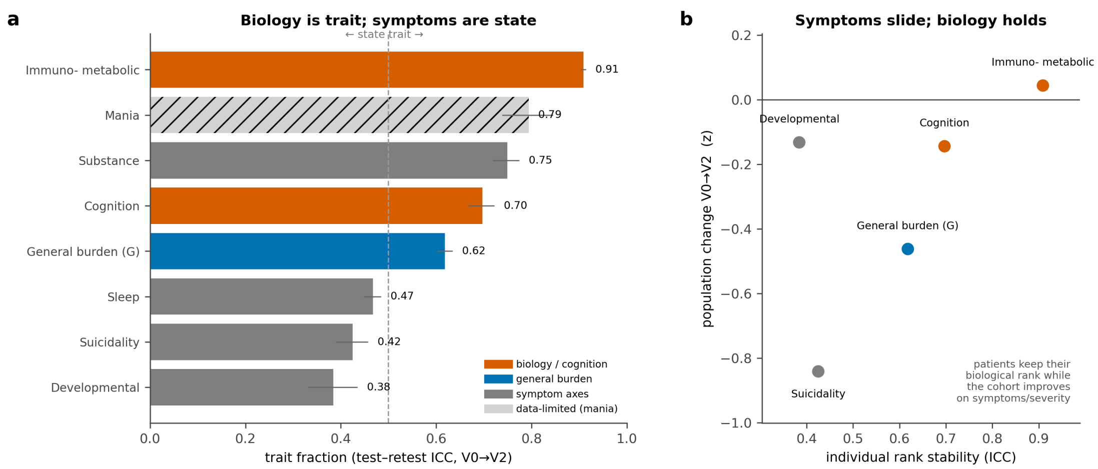


Figure 4: **Biology is trait; symptoms are state.** (a) Trait fraction (test–retest ICC, baseline→2 years) for each axis; immunometabolic load is the most durable (0.91). (b) Individual rank stability against population change: symptom axes slide as the cohort improves, while immunometabolic load holds its rank (upper right).

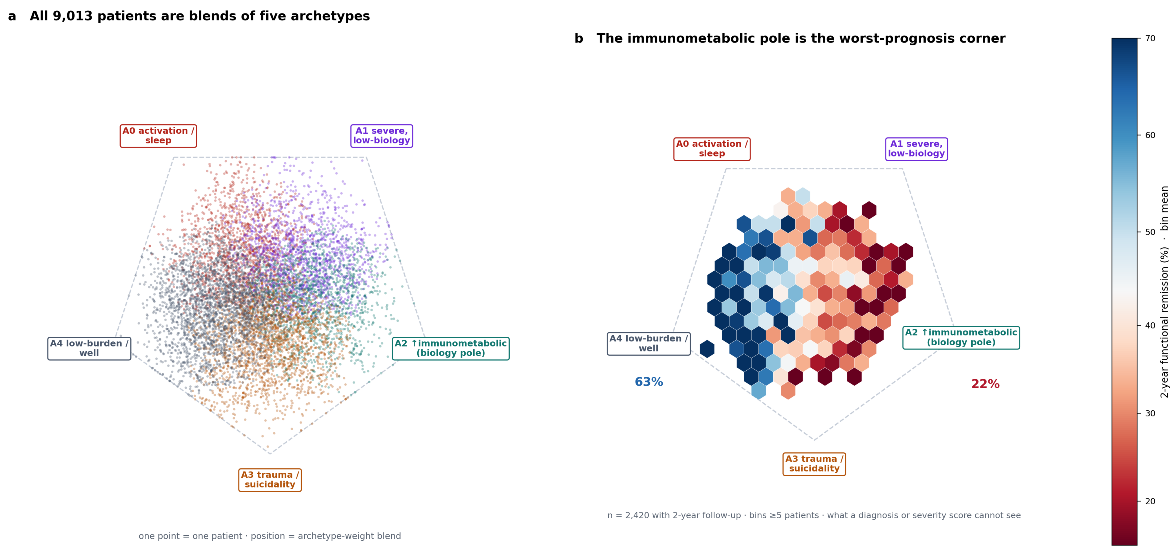


Figure 5: **The immunometabolic pole is the worst-prognosis corner.** (a) All 9,013 patients projected onto the five-archetype simplex as convex blends; position is the archetype-weight mix. (b) The same map shaded by two-year functional remission (bin mean, bins $\geq 5$ patients, $n = 2{,}420$ with follow-up): remission falls from 63% at the well pole to 22% at the immunometabolic pole. The immunometabolic corner is worse than the equally severe clean-biology corner—an axis a diagnosis or a severity score does not resolve.

## 2.5 It marks the worst-prognosis pole—at the group level

Finally we asked whether the map carries prognosis. Projecting all 9,013 patients onto the five-archetype simplex and shading by two-year functional remission reveals a clean gradient across the patient space, from the low-burden "well" pole (remission 63%) to the **immunometabolic pole** (remission 22%), which is the worst-prognosis corner of the map (Figure 5). Decisively, the immunometabolic corner (22%) is worse than the *equally high-severity* clean-biology corner (34%): the two poles carry the same general burden but diverge in outcome, so a severity score—which cannot tell them apart—misses the gap. The remission gradient holds *within* every diagnosis, indicating it is a property of the map rather than of the diagnostic mix (Figure 6).

We are deliberate about the size of this effect. An errors-in-variables Bayesian generalized linear model, which propagates each patient's factor-score uncertainty into the outcome regression, shows the archetype encoding improves prediction of two-year functioning beyond diagnosis, severity and baseline outcome. On held-out patients the gain in expected log predictive density—a standard measure of how much better a model forecasts unseen outcomes, larger is better—is $\Delta$ELPD $= +62.8$, a clear group-level improvement; but the individual-level discrimination gain is small (AUC +0.010, barely above the +0 of no improvement). On the same scale the map adds +17.3 where the diagnosis label adds +29.0, so the two are co-informative—the map carries prognostic information of the same order as diagnosis, not exceeding it. The prognostic signal is real and group-level; it is not, on this observational baseline cohort, an individual-level predictor.

Across the four tests, one axis answers to the same name each time: immunometabolic load is the *least* entangled with general severity (correlation $\approx 0.10$ versus 0.39/0.42), the *most* durable over two years (ICC 0.91), and the *worst*-prognosis pole (22% versus 63% functional remission). No single one of these numbers is the result; their convergence on one axis is. That convergence positions the immunometabolic axis as a cohort-enrichment and monitoring target—a durable, severity-independent axis that concentrates poor outcomes—rather than as an individual treatment rule.

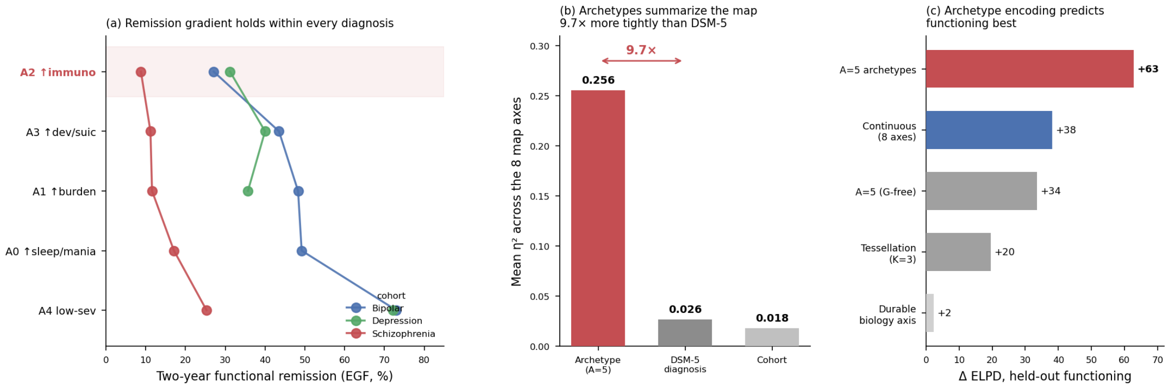


Figure 6: **The remission gradient is a property of the map.** (a) The two-year functional-remission gradient across archetypes holds within each diagnosis (bipolar, schizophrenia, depression scored separately). (b) Archetypes summarize the map $\approx 9.7\times$ more tightly than the DSM-5 label ($\eta^2$ across the eight axes). (c) Incremental held-out predictive density for functioning: the archetype encoding adds most, above the continuous map, tessellation and durable-biology-only encodings.

# 3 Discussion

Across four validation stages one property of the map recurs. Immunometabolic load is the axis that stands apart from general severity ($\phi_{Gd} \approx 0.10$ versus 0.39/0.42 for cognition and sleep), the axis that stays put over two years (test–retest ICC 0.91), and the axis that marks the worst-prognosis pole of the patient space (two-year functional remission 22% at the immunometabolic corner versus 63% at the well pole). No single one of these observations would carry the argument; their convergence—separable *and* durable *and* prognostic, the same axis each time—is what identifies the immunometabolic dimension as a distinct, stable, outcome-relevant feature of severe mental illness rather than another face of being severely ill.

This headline is corroborated, in direction, by a large and independent literature, which is why we state it as *relative* rather than absolute independence. Metabolic burden in severe mental illness is driven by adiposity, medication, age and chronicity more than by diagnosis or symptom severity, is present transdiagnostically across schizophrenia, bipolar disorder and depression [9–11, 16, 17]; inflammation marks a subgroup and specific symptoms rather than the severity continuum [18–20]; and the immunometabolic-depression programme has independently arrived at a separable biological dimension mapping onto atypical, energy-related symptoms [12–15, 21]. Our contribution is not to re-discover this biology but to *measure* it inside one diagnosis-blind map: to give the qualitatively-known separation a number ($\approx 1\%$ shared variance with general burden), estimated with a model honest enough—no imputation, uncertainty carried end-to-end, the map frozen before validation—that the number can be trusted.

The practical reading is a design principle, not a treatment rule. The axes divide cleanly into those worth *stratifying* on and those worth *monitoring*: a durable, severity-independent axis that concentrates poor outcomes is what one enriches a trial or a service cohort on, whereas the mobile symptom axes are what one tracks over time. The immunometabolic axis sits squarely on the stratify side—it is still there at two years, it is not captured by a severity score, and it identifies a corner of the map where only one patient in five reaches functional remission. That makes it a concrete enrichment and monitoring target: a rationale for measuring and following routine cardiometabolic and inflammatory biology as an axis of illness in its own right, and for testing—prospectively—whether intervening on it changes course.

**Limitations.** Three classes bound these claims. *First, the results are observational and internal.* Every finding is an internal-validity result on baseline and two-year data from a single consortium; the map's structure, separability and prognostic gradient are estimated, not externally validated, and causal language is deliberately avoided. *Second, the prognostic signal is group-level.* The archetype encoding improves held-out predictive density for two-year functioning ($\Delta$ELPD = +62.8) but adds little individual-level discrimination (AUC +0.010) and is co-informative with, not superior to, diagnosis. The immunometabolic pole is an enrichment target, not an individual predictor, and we report the modest effect as plainly as the strong ones. *Third, coverage and horizon are finite.* Follow-up outcomes were available for 2,420 of 9,013 patients, the biology panel is the routine one collected across these cohorts rather than a deep immunophenotype, and two years is a short window for functional trajectories.

Establishing individual-level clinical utility would require external validation in independent cohorts, incident-event outcomes over longer horizons, and—for any causal claim about the immunometabolic axis—prospective or interventional data beyond this baseline observational cohort. What the present map establishes is prior to all of that and necessary for it: that when biology is measured honestly and placed inside the transdiagnostic space as a co-equal axis, one biological dimension emerges as distinct, durable and prognostic—an immunometabolic island that a diagnosis or a severity score cannot see.

# 4 Methods

## 4.1 Cohorts, harmonization and the no-imputation data layer

We analysed the harmonized baseline assessment of $N = 9{,}013$ patients with bipolar disorder (6,252), schizophrenia (2,209) or major depression (552) from three national FACE cohorts across 21 expert centres. Instruments common to the cohorts were mapped to a shared 143-variable pool spanning cognition, symptom and functioning scales, sleep and chronotype, developmental and substance-use history, and a routine biology panel (body composition, lipids, glycaemia, urate, C-reactive protein). Missingness is pervasive and non-random in such registries; rather than impute, we treat each patient's observed cells as the likelihood target and marginalize the rest, so no cell is ever filled in. Diagnosis was withheld from model fitting throughout and used only to describe, never to define, the map.

## 4.2 The measurement model: a Bayesian bifactor structure

The core is a single global Bayesian sparse bifactor / exploratory structural-equation model [22, 23]. The generative hypothesis is that a small set of latent axes is the *only* reason a patient's many indicators move together: each patient $i$ carries an eight-dimensional latent score

$$f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \ \sim\ \mathcal{N}_{K+1}(\mathbf{0}, \Phi)\,, \qquad \mathrm{diag}(\Phi) = \mathbf{1}, \tag{1}$$

a general functional-burden factor $G$ and $K = 7$ specific axes (cognition, immunometabolic load, sleep, mania/activation, suicidality, developmental risk, substance use). $\Phi$ is a *correlation* matrix: its unit diagonal fixes the otherwise-arbitrary latent scale, so a factor's variance is normalized to one and only the loadings carry magnitude. Every indicator $j$ is a linear reflection of these factors through a single linear predictor

$$\eta_{ij} = \alpha_j + \lambda_{jG}\, G_i + \sum_{k=1}^{K} \lambda_{jk}\, D_{ik} + \beta_j^\top c_i, \tag{2}$$

with item intercept $\alpha_j$, a *general* loading $\lambda_{jG}$ on $G$, *specific* loadings $\lambda_{jk}$ (row $j$ of the loading matrix $\Lambda = [\lambda_G \mid \Lambda_D]$), and item-level covariates $c_i$ that shift an indicator's mean (age, sex, site) but are never themselves factor indicators. The term *bifactor* is exactly this structure: every indicator may load on $G$ *and* on its home specific factor, so general burden and a domain-specific signal are separated within each item rather than confounded. It is *exploratory* rather than confirmatory because an indicator is *allowed* small cross-loadings on plausibly related factors, governed by a regularized-horseshoe prior [24, 25] that shrinks the cross-loading matrix toward simple structure while leaving anchored home-factor loadings free—so sparsity is chosen by the data rather than imposed.

**Variance decomposition.** Taking the Gaussian indicators as the illustrative case (the mixed-likelihood generalization follows below) and stacking one patient's items, $x_i = \mu_i + \lambda_G G_i + \Lambda_D D_i + \varepsilon_i$—with mean term $\mu_i = \alpha + B c_i$ collecting the intercepts and covariate shifts of Eq. 2, and residual $\varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi)$, diagonal $\Psi = \mathrm{diag}(\sigma_1^2, \dots, \sigma_J^2)$—each indicator's variance splits into three interpretable parts,

$$\mathrm{Var}(x_{ij}) = \underbrace{\lambda_{jG}^2}_{\text{general}} + \underbrace{\textstyle\sum_{k=1}^{K} \lambda_{jk}^2}_{\text{specific}} + \underbrace{\sigma_j^2}_{\text{unique}}\,, \tag{3}$$

the share attributable to overall burden, to the domain axes, and to item-specific noise. The *defining hypothesis* that makes this decomposition valid is **local independence**: given a patient's latent scores the items are conditionally independent, $p(x_i \mid f_i, \Theta) = \prod_{j \in O_i} p(x_{ij} \mid f_i, \Theta)$, i.e. the factors explain away all shared covariance. In the linear-Gaussian case this is *identical* to "$\Psi$ is diagonal." This is also why missing cells are marginalized, never imputed: an invented value would inject between-cell covariance that the model would mistake for a factor.

**Observation model and identification.** Indicators are generated through mixed likelihoods appropriate to each variable's type—Gaussian for continuous and log-transformed laboratory values, ordinal-logit for scales, Bernoulli for binary, negative-binomial for counts—with a Gaussian-copula block coupling the correlated continuous biology panel [26], so no variable is coerced onto a foreign scale. Because $f_i$ is unobserved it is integrated out, and fitting targets each patient's observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \psi_j)\,, \tag{4}$$

where $\Theta$ collects all model parameters, $O_i$ is the set of indicators observed for patient $i$, $F_j$ is the per-indicator likelihood chosen by variable type (above) and $\psi_j$ its dispersion/threshold parameters. Unobserved indicators are marginalized via the Woodbury identity for the copula block—which avoids repeatedly inverting a large patient-specific covariance submatrix and is the no-imputation invariant on which every downstream analysis rests. The bifactor identification fixes $G$ orthogonal to the specific axes ($\Phi$ block-diagonal); we treat this as an identifying coordinate choice defining the *primary map*, not an empirical claim about independence.

## 4.3 The correlated-$G$ arm and the primary estimand

To measure—rather than assume—whether biology is separable from severity, we re-estimated the model with the block-diagonal constraint released, allowing $G$ to correlate freely with the specific axes:

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^{\top} \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \qquad \phi_{Gd} = \operatorname{Corr}(G, D) \in [-1, 1]^7. \tag{5}$$

The vector $\phi_{Gd}$ (Eq. 5) is the *primary estimand* for the biology–severity question; the bifactor map remains the coordinate system for all other analyses. Robustness of the immunometabolic separation was assessed by re-estimating $\phi_{Gd}$ after adjusting each indicator for age, sex, education and recruitment site, and then additionally for antipsychotic exposure.

## 4.4 Estimation, invariance and scoring

Posteriors were sampled with the No-U-Turn Sampler [27] in a probabilistic-programming implementation [28, 29], with a fast stochastic-variational re-estimation [30–32] used to confirm the atlas at reduced cost (variational and sampled factor scores agree to within posterior uncertainty). Key priors are weakly-informative: a regularized-horseshoe on the cross-loadings (global scale $\tau_0 = 0.05$, slab $c = 0.30$) and $\Phi \sim \text{LKJ}(\eta{=}2)$. Intermediate stages advanced only under a strict gate ($\hat{R} \leq 1.01$, effective sample size $\geq 400$, zero divergences); the reported joint eight-factor (mixed-likelihood) map is taken at the largest sample that mixes ($\hat{R} \leq 1.03$, zero divergences), with a cross-seed stability guard (Tucker congruence $\varphi = 0.99$). Measurement invariance was checked across the three cohorts (Extended Data Fig. E3) and by leave-one-cohort-out refitting (Extended Data Fig. E4), with split-half reproducibility benchmarking (Extended Data Fig. E5); patient factor scores and their posterior uncertainty were obtained under the frozen map. All validation analyses below were pre-specified against the fixed map; the map was never re-fit to improve a downstream result.

### 4.5 Structure testing and archetypes

Whether the patient space is clustered or continuous was tested by comparing the best silhouette (a standard cluster-separation score: within- versus between-cluster distance) over $K = 2$–12 clusters against the distribution expected under a single-multivariate-Gaussian null (a parametric bootstrap of the same covariance), and by Gaussian-mixture model selection. The continuum was summarized by archetypal analysis [33], which represents each patient as a convex blend of extreme phenotypes (weights $w_a \geq 0$, $\sum_a w_a = 1$); across candidate counts the five-archetype solution was retained as the point beyond which reconstruction error flattened and cross-seed archetype recovery remained stable. Descriptive tightness was quantified as the mean $\eta^2$ (the proportion of between-patient variance in a factor score explained by an encoding—unrelated to the linear predictor $\eta_{ij}$ above) of each encoding (archetype, DSM-5, cohort) across the eight axes: a higher value means the encoding localizes a patient more tightly on the map.

### 4.6 Temporal coherence and trait/state

Two yearly follow-ups were scored under the frozen map. A between/within/measurement variance decomposition yielded a test–retest intraclass correlation per axis,

$$\mathrm{ICC} = \frac{\tau^2}{\tau^2 + \omega^2}, \tag{6}$$

with $\tau^2$ the durable between-patient (trait) variance and $\omega^2$ the within-patient plus measurement (state) variance, separating durable traits from moving states.

### 4.7 Prognosis: errors-in-variables incremental validity

Two-year functional remission (based on the EGF, the French Global Assessment of Functioning scale) and continuous functioning were modelled with an errors-in-variables Bayesian generalized linear model that propagates each patient's factor-score uncertainty into the outcome regression,

$$\theta_{id} \sim \mathcal{N}(\hat{f}_{id}, s_{id}^2), \qquad g(\mathbb{E}[y_i]) = \beta_0 + \beta_W^\top W_i + \beta_\theta^\top \theta_i, \tag{7}$$

with $\hat{f}_{id}$ the posterior factor score, $s_{id}$ its posterior SD and $W_i$ the covariates (diagnosis, baseline severity, baseline outcome), avoiding regression dilution from treating latent scores as known. Incremental validity was assessed by expected log pointwise predictive density on held-out data [34, 35], with individual-level discrimination (AUC) reported alongside. Because retention at two years is associated with baseline map position, the incremental-validity estimate was confirmed under inverse-probability-of-attrition weighting. The archetype-prognosis map (Fig. 5) bins patients by their archetype-weight position and shades by mean two-year functional remission ($n = 2{,}420$ with follow-up, bins $\geq 5$ patients).

### 4.8 Data and code availability

Individual patient data are governed by the FACE cohort agreements and are not publicly redistributable; derived aggregate outputs underlying the figures, and the modelling and figure-generation code, are available from the corresponding author on reasonable request. Per-patient data are never required to regenerate the aggregate results.

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# Extended Data

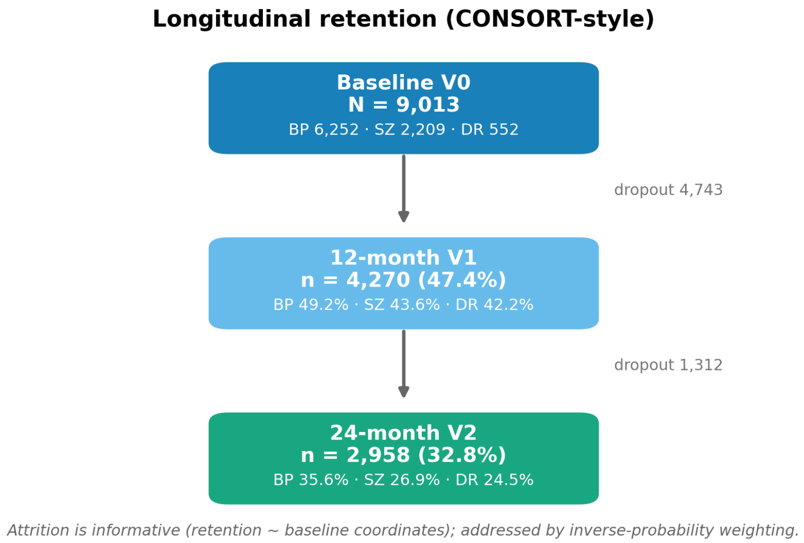


Figure E1: **Cohort assembly and follow-up attrition.** Flow from the three FACE cohorts to the $N = 9{,}013$ modelled patients and the $n = 2{,}958$ retained at 24 months (of whom $n = 2{,}420$ have a complete two-year functional-remission outcome, the sample used in Figure 5), with reasons for exclusion at each stage. Retention is associated with baseline map position and is addressed by inverse-probability weighting.

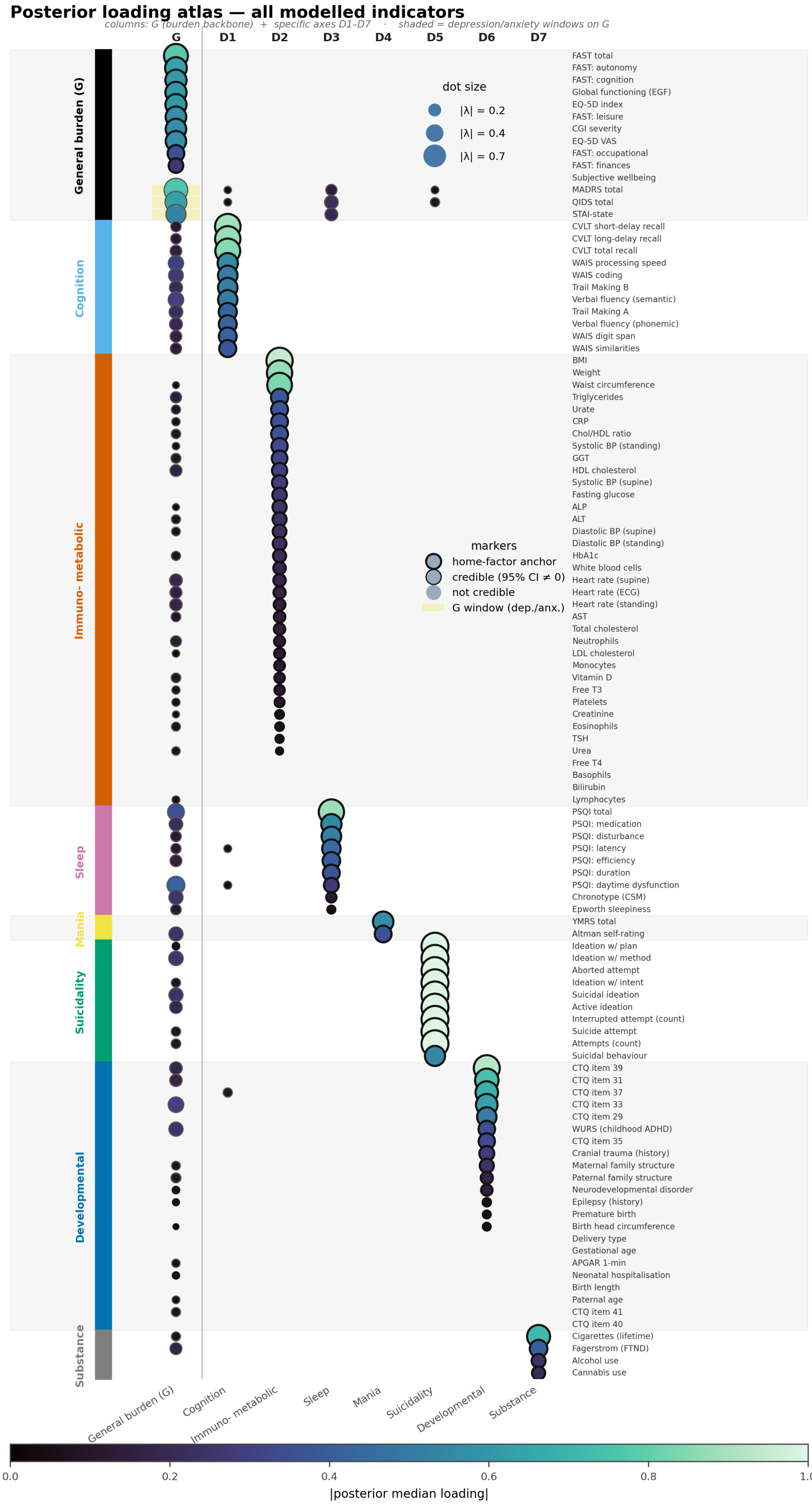


Figure E2: **The full eight-axis loading atlas.** Complete posterior indicator×factor loading matrix for all 143 modelled indicators, extending the summary of Figure 1 to every variable and both the bifactor and correlated-$G$ parameterizations.

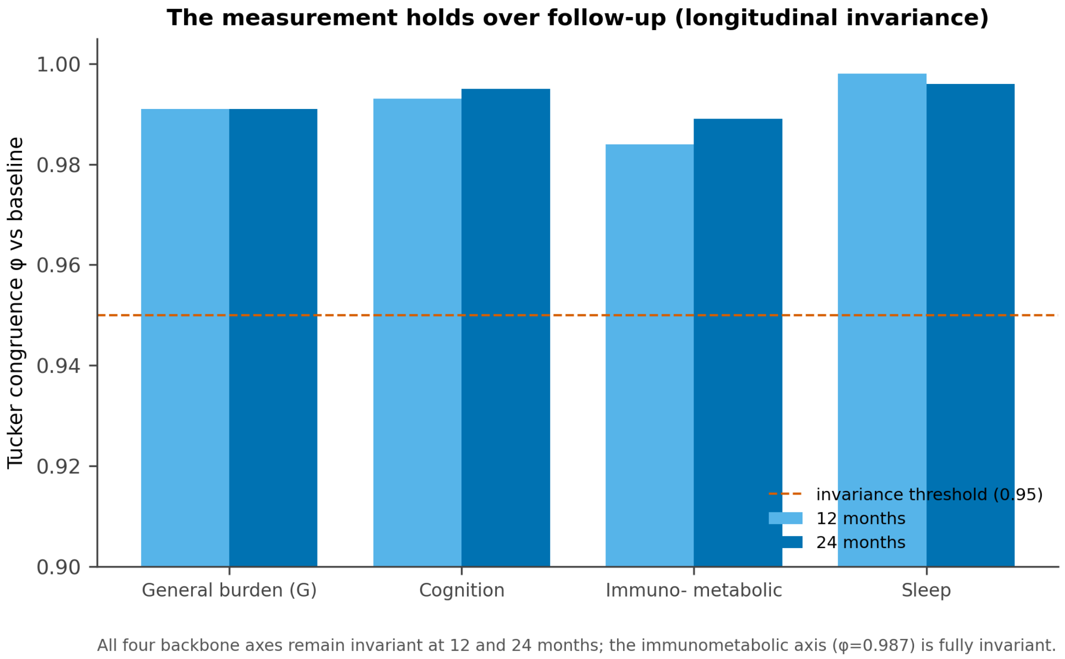


Figure E3: **Measurement invariance across cohorts.** Configural, metric and scalar invariance checks for the eight-axis map across the three cohorts; loadings and thresholds are stable enough that factor scores are comparable across sites.

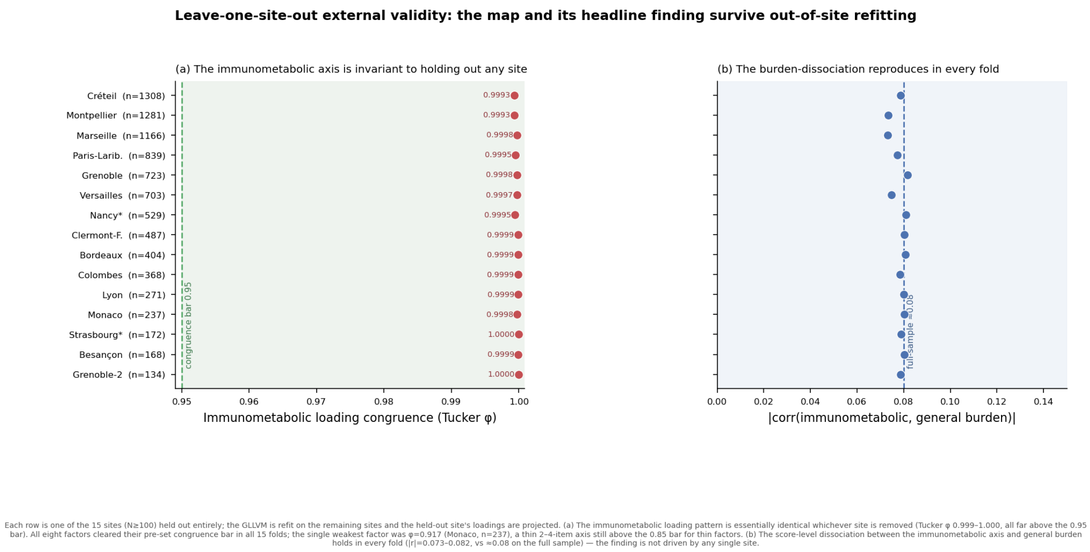


Figure E4: **Leave-one-cohort-out stability.** The map re-estimated with each cohort held out in turn; axis definitions, the immunometabolic separation and the archetype geometry are reproduced, showing no single cohort drives the structure.

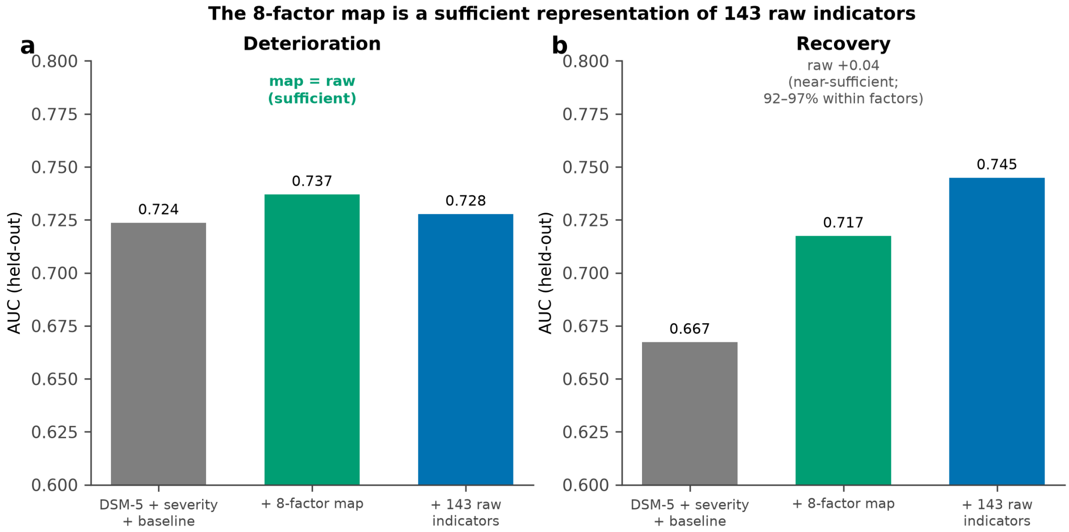


Figure E5: **Reproducibility benchmarking.** Split-half and re-fit reproducibility of the loading atlas and the general–specific correlation vector $\phi_{Gd}$, with the immunometabolic axis among the most reproducible.

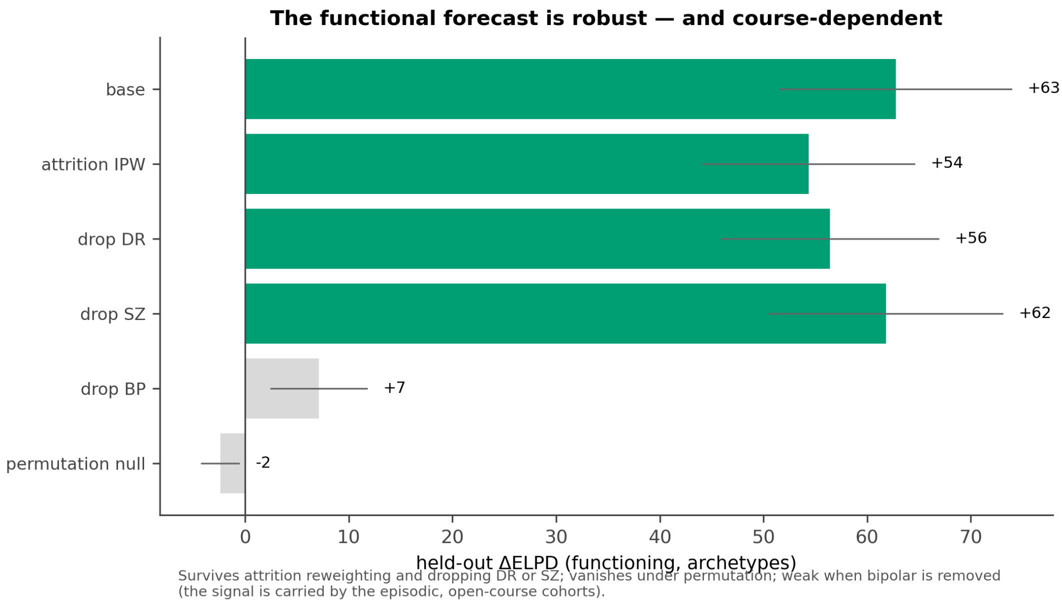


Figure E6: **Confound robustness of the biology axis.** The immunometabolic–$G$ correlation $\phi_{Gd}$ under progressive adjustment (age, sex, education, site, then antipsychotic exposure): it does not rise with adjustment, falling from $\approx 0.10$ to $\approx 0.07$, and remains the least burden-entangled axis at every step.